DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 25: The Black–Scholes–Merton PDE

Delta-Hedging · The Riskless Portfolio · Why μ Disappears

October 11, 2026

Today's Agenda

1. The setup: everything we already have

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2. The dynamics of the derivative
3. The delta-hedged portfolio
4. No-arbitrage $\Rightarrow$ the PDE
5. Why μ vanishes

Learning Objectives

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- Derive the BSM PDE from Itô + no-arbitrage.
- Explain why μ does not appear.
- State the boundary conditions.

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The task: combine them.

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Week 5: build a stock+bond portfolio replicating the payoff, invoke the Law of One Price. Today: the tree has become a continuum – neutralize randomness *instant by instant*. Same mechanism (Δ shares against the option), but Δ adjusts continuously. The reward: one PDE that prices *every* derivative at once.

Step 1: Itô on $V(S,t)$

With drift $a = \mu S$, diffusion $b = \sigma S$:

$$\mathrm{d}V = \left(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)\mathrm{d}t + \sigma S\, V_S\, \mathrm{d}W_t$$

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The Choice That Kills the Randomness

$$\Delta = \frac{\partial V}{\partial S} \qquad \Longrightarrow \qquad \mathrm{d}\Pi = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)\mathrm{d}t$$

Instantaneously riskless.

Look What Happened to μ

Remark

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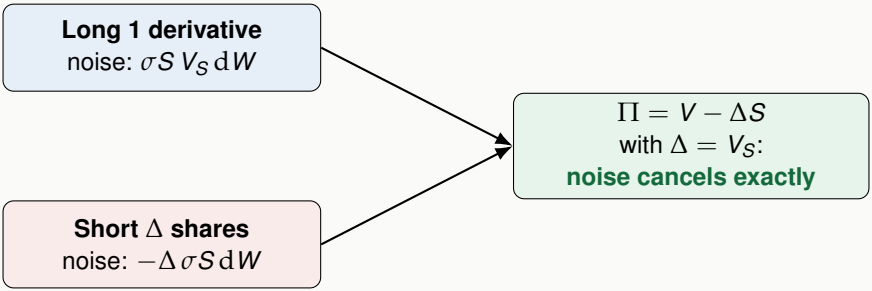
Setting $\Delta = V_S$ kills the $\mathrm{d}W$ term. But look at the $\mathrm{d}t$ term: $\mu S V_S$ and $-\Delta \mu S$ **cancel simultaneously**. The drift disappears as a *free by-product* of hedging away the risk. Not a coincidence – the single most important structural fact in the derivation.

The Cancellation, Pictured

Long 1 derivative
noise: $\sigma S\, V_S\, \mathrm{d}W$

Short Δ shares
noise: $-\Delta\, \sigma S\, \mathrm{d}W$

The Cancellation, Pictured



Step 3: A Riskless Portfolio Must Earn r

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The Black–Scholes–Merton PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

Why the Equality Is Forced

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If the riskless portfolio earned *more* than r: borrow at r, buy it, pocket the difference with certainty and zero net investment. A textbook Type A arbitrage (Week 5). If *less*: reverse the trade. So $\mathrm{d}\Pi = r\Pi\,\mathrm{d}t$ is forced, not assumed.

The Central Surprise

Key Takeaway

The PDE contains σ, r, S, t – but **not** μ.

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Key Takeaway

The PDE contains σ, r, S, t – but **not** μ. Two investors who disagree completely about a stock's expected return must agree *exactly* on every option price.

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Destination vs. Erratic-ness

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And Week 8 showed μ is the parameter we cannot estimate. Spectacular luck.

Confirmed Numerically

400,000 paths, $S_0=100$, $K=105$, $r=5\%$, $\sigma=20\%$, $T=1$

Real-world μ	-20%	0%	$+10\%$	$+50\%$

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Expected terminal price ranges 82 **to** 165**; every option price is** 8.02**.**

One Equation, Many Contracts

Key Takeaway

- **Call**: $V(S,T) = \max(S-K,0)$, $V(0,t) = 0$, $V \to S - Ke^{-r(T-t)}$ as $S \to \infty$.

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- **Call**: $V(S,T) = \max(S-K,0)$, $V(0,t) = 0$, $V \to S - Ke^{-r(T-t)}$ as $S \to \infty$.
- **Put**: $V(S,T) = \max(K-S,0)$, $V(0,t) = Ke^{-r(T-t)}$, $V \to 0$ as $S \to \infty$.

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A Law of Physics Plus an Initial Profile

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With $x = \ln S$, reversed time, and an exponential rescaling, the BSM PDE **becomes** the heat equation $u_\tau = u_{xx}$ – solved since Fourier.

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With $x = \ln S$, reversed time, and an exponential rescaling, the BSM PDE **becomes** the heat equation $u_\tau = u_{xx}$ – solved since Fourier. That is how Black and Scholes could write down an *answer*, not just an equation.

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- μ is absent: disagree about returns, still agree on option prices.
- The payoff selects which derivative the PDE is describing.

Check Your Understanding

Discuss (2 minutes)

1. Verify $V(S,t) = S$ satisfies the PDE. What is Δ?

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1. Verify $V(S,t) = S$ satisfies the PDE. What is Δ?
2. Verify $V(S,t) = Ke^{-r(T-t)}$ satisfies it.
3. We held Δ fixed over $\mathrm{d}t$ while setting $\Delta = V_S$, which itself moves. Why is this OK as $\mathrm{d}t \to 0$?

Looking Ahead

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The Black–Scholes Formula – Lecture 26

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The Black–Scholes Formula – Lecture 26

And confirm it's the limit your binomial trees were approaching.